TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Paul Mentink (M.Sc.), Senior Research Scientist, Powertrains	**Email**	paul.mentink@tno.nl
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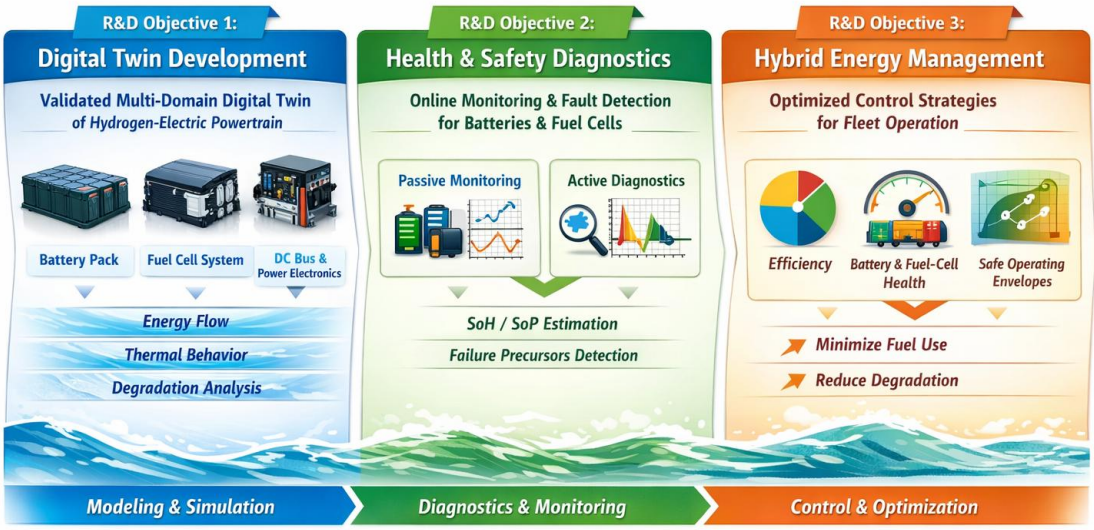
2. Technology Overview

Title of Proposed Technology	Digital Twin and Safety Diagnostics for Hydrogen-Electric Maritime Powertrains			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Small Category	Medium Category	Medium Category
	Batteries	Secondary Batteries	Software	SW Solutions
Concept of Proposed Technology	Hydrogen-electric vessels combine lithium-ion battery packs, power electronics and hydrogen fuel-cell systems to meet zero-emission targets while maintaining safety and availability. This proposal develops an integrated, control-oriented digital twin for the powertrain, coupled with advanced battery and fuel-cell diagnostics.			

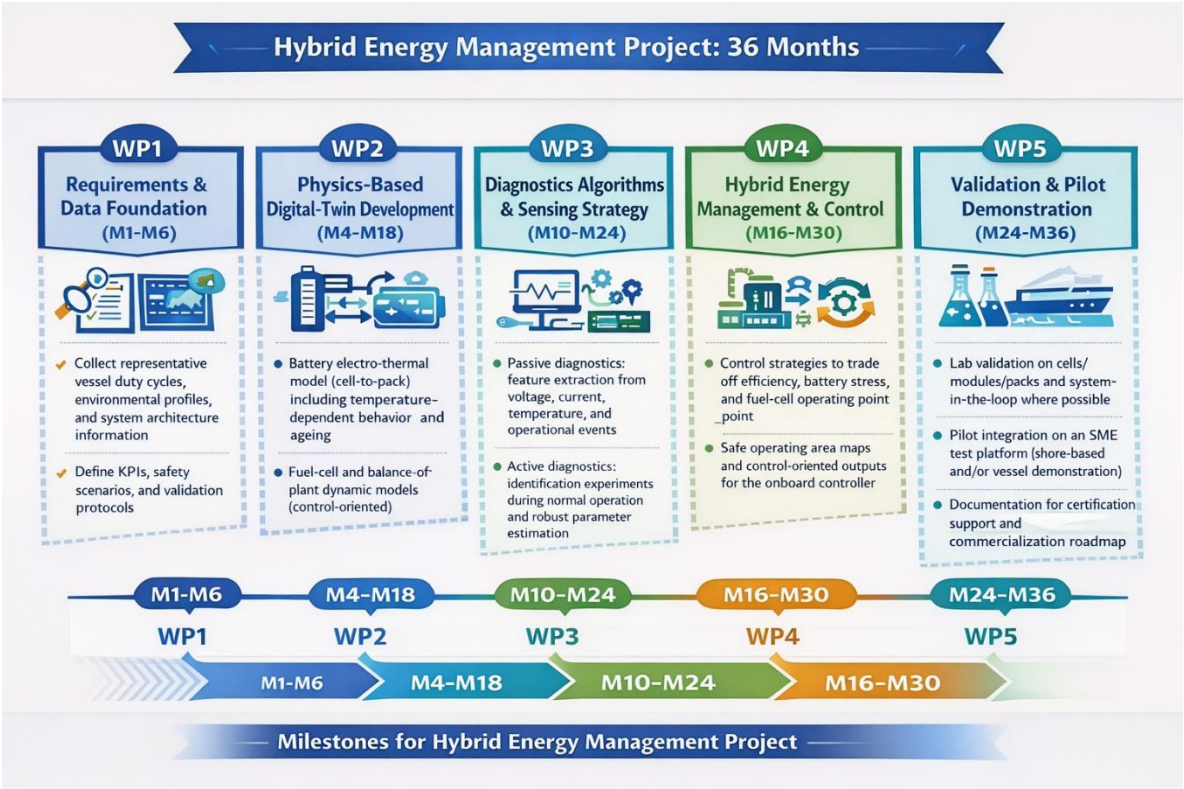
	   Core elements: • Physics-based electro-thermal battery models (cell-to-pack) including ageing and temperature effects. • Online diagnostics using voltage/current/temperature plus active system identification and electrochemical impedance spectroscopy (EIS) where feasible. • Hybrid energy management and optimal control to minimize fuel use and degradation under maritime duty cycles. • A validation workflow using lab testing (cells, modules, and packs) and hardware-in-the-loop to de-risk onboard deployment. Intended use: design optimization, safety monitoring and predictive maintenance for maritime hybrid systems (battery + fuel cell), supporting faster certification and reduced lifecycle cost.
Rationale for Joint R&D Project	Maritime hydrogen-electric systems face accelerated degradation from high-power transients, variable ambient conditions, and strict safety requirements. Korean SMEs developing such systems need validated models and diagnostic methods to (i) quantify battery and fuel-cell health in operation, (ii) detect early failure precursors, and (iii) optimize energy management to extend lifetime. Collaboration is required because: • TNO provides proven methodologies for model-based energy management, battery digital-twin development, and online diagnostic approaches (including active diagnostics

	and EIS-based methods) documented in peer-reviewed publications. • The Korean SME provides system design details, operational profiles, and a pilot platform for integration and demonstration. Together, we can deliver a validated digital-twin and safety diagnostics toolkit ready for real-world trials under the TIPA program.
Key Words	1. Maritime powertrain digital twin 2. Battery diagnostics (active and passive) 3. Electrochemical impedance spectroscopy (EIS) 4. Hybrid energy management (battery + fuel cell) 5. Predictive maintenance and safety monitoring

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4-5)	End of the Project	Level (6-7)
R&D Objectives	 R&D Objective 1: Develop a validated, multi-domain digital twin of a maritime hydrogen-electric powertrain (battery pack, fuel-cell system, DC bus, and key auxiliaries) that captures energy flow, thermal behavior, and degradation drivers under representative duty cycles. R&D Objective 2: Implement online health and safety diagnostics for the battery and fuel-cell system, combining passive monitoring with active system identification (and EIS where feasible), to estimate SoH/SoP and detect failure precursors. R&D Objective 3: Design hybrid energy management and control strategies that minimize fuel consumption and degradation, providing safe operating envelopes and actionable KPIs for fleet operation..			
R&D Target Specifications	**SoH estimation accuracy:** less than or equal to 2 percent error versus reference characterization. **Early anomaly detection:** detect abnormal behavior at least 30 minutes before critical safety thresholds for defined safety scenarios. **Lifetime improvement through control:** at least 15 percent reduction in battery degradation on agreed duty profiles. **Demonstration maturity:** integrated validation in a representative test setup and pilot environment, supporting TRL progression from 4 to 5 at project start to 6 to 7 at project end.			

R&D Plan



Work Package 1 (Months 1-6): Requirements and data foundation

• Collect representative vessel duty cycles, environmental profiles, and system architecture information.

• Define KPIs, safety scenarios, and validation protocols.

Work Package 2 (Months 4-18): Physics-based digital-twin development

• Battery electro-thermal model (cell-to-pack) including temperature-dependent behavior and ageing.

• Fuel-cell and balance-of-plant dynamic models (control-oriented).

• Coupled system model for DC bus and power electronics (efficiency and limits).

Work Package 3 (Months 10-24): Diagnostics algorithms and sensing strategy

• Passive diagnostics: feature extraction from voltage, current, temperature, and operational events.

• Active diagnostics: identification experiments during normal operation and robust parameter estimation.

• EIS-based methods for high-voltage systems, with safe excitation and measurement procedures.

Work Package 4 (Months 16-30): Hybrid energy management and control

• Control strategies to trade off efficiency, battery stress, and fuel-cell operating point.

	• Safe operating area maps and control-oriented outputs for the onboard controller. Work Package 5 (Months 24-36): Validation and pilot demonstration • Lab validation on cells/modules/packs and system-in-the-loop where possible. • Pilot integration on an SME test platform (shore-based and/or vessel demonstration). • Documentation for certification support and commercialization roadmap.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) will lead the R&D on modeling, diagnostics and validation: • System architecture and digital-twin development (battery, fuel cell, thermal, and power electronics). • Development of online diagnostics methodologies (passive plus active identification), including EIS-based approaches. • Development of hybrid energy management and control strategies. • Validation design, test execution in TNO facilities (cell/module testing, climatic testing where applicable), and model verification. • Dissemination via publications and jointly agreed IP strategy with the Korean SME.
Expected R&R of Korean SME Partner	Korean SME Partner will lead system integration and demonstration: • Provide maritime hydrogen-electric powertrain architecture, constraints, and representative operational profiles. • Provide relevant datasets (BMS, fuel-cell controller signals, environmental conditions) and support instrumentation. • Support definition of safety scenarios, failure modes, and acceptance criteria. • Integrate algorithms/digital-twin components into their test platform and support pilot demonstration. • Lead commercialization planning, customer engagement, and regulatory/certification alignment for maritime deployment.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical outcomes: • A validated digital-twin toolkit for hydrogen-electric maritime powertrains, enabling design-space exploration and safer operation. • Battery and fuel-cell online diagnostics algorithms (SoH/SoP estimation, anomaly detection) with validated performance metrics. • Hybrid energy management and control strategies providing measurable reductions in fuel use and degradation. Impact: • Improved safety and reliability for zero-emission vessels, supporting market entry and certification. • Reduced lifecycle cost through predictive maintenance and optimized operating envelopes. • Strengthened Korea-EU collaboration capability under TIPA, with potential follow-on pilots and standardization activities.

6. References

(1) Wilkins, S., Rosca, B., Jacob, J., Hoedmaekers, E. Optimised battery capacity utilisation within battery management systems. EVER 2015 (IEEE), 2015. TNO Repository entry: https://repository.tno.nl/SingleDoc?docId=42569 (accessed 2025-12-18).

(2) Rath, S., Medina, R., Wilkins, S. Real-time optimal charging strategy for a fleet of electric vehicles minimizing battery degradation. IEEE, 2023. PDF: https://publications.tno.nl/publication/34641555/EemswK/rath-2023-real.pdf (accessed 2025-12-18).

(3) Dadikozyan, A., Beckers, C., Meulenbroeks, T., van den Tillaart, E., Wilkins, S. Battery Diagnostics and Monitoring Methods - A Comparative Analysis of Active versus Passive Approaches. EVTeC 2025 preprint, April 2025. ResearchGate: https://www.researchgate.net/publication/390835686 (accessed 2025-12-18).

(4) Xie, N., Zhao, W., Lin, W., Chen, Y. A Method for Measuring Electrochemical Impedance Spectroscopy in High-voltage Battery Systems. 2024 International Conference on Energy and Electrical Engineering (EEE), July 2024. DOI: https://doi.org/10.1109/EEE59956.2024.10709725 (metadata page: https://www.researchgate.net/publication/384980187, accessed 2025-12-18).

(5) TNO. Modular Energy Management Strategy (MEMS). Web page. https://www.tno.nl/en/sustainable/mobility-logistics/energy-management/ (accessed 2025-12-18).

☐ Terms:

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization